

AC outputs use G3MB-202P DC5 SSRs: 2 A max per output; AC load rating: R1 Datasheet load range is 75–264 VAC; 24 VAC use is intentionally out-of-datasheet/prototype – Board current assumption: 1 oz copper, 2.0 mm AC tracks. Limit shared solenoid common bus to 3 A total continuous; each switched output and pump output to 2 A max. Fusing required: SOLL common fuse <= 3 A, each load must be externally protected for wiring/load. 5V fuse <= 1 A.

Clearance policy: AC_HIGH_CURRENT nets use 1.5 mm clearance to match 3.81 mm terminal pitch. This is not certified mains insulation; final product needs EE/safety review.

SSR drive: ESP32 GPIO drives NPN low-side transistor via 4.7k. Firmware must configure outputs LOW before enabling loads. Add hardware base pullups in next rev if needed.

Low-voltage I/O: ESP32 GPIO only, 3.3 V logic, 12 mA max recommended per pin. Boiler probe divider is 10K/10K; probe input <= 5V nominal, <= 6.6 V absolute to LV pins.

